CS315 · ALGORITHMS ANALYSIS & DESIGN

What Repeats

An analysis of three midterms: 472, 451 and 442

Muharib

What's inside

01 The bottom line

02 The repetition table

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01 The bottom line

I compared the three midterms I received. The result is clear: the 472 and 451 midterms are almost the same paper, down to the numbers.

Question	451	472
Matrix chain	A=4×5, B=5×8, C=8×3, D=3×7	Exactly the same.
LCS	X = ACBAED, Y = ABCADF	Exactly the same.
Hash table	[4, 13, 49, 6, 40, 33], size 9	The same keys, chaining only.
B-tree	25, 53, 91, 41, 12, 87, 62, 51	Every number minus 1.
True / False	5 items.	4 of them repeated.

So

If you solve the 451 midterm and understand every question in it, you have covered most of the midterm's shape. Do not memorise the numbers, though. Learn the method, because the instructor may change them.

02 The repetition table

Topic	472	451	442	Lecture
Counting operations of **arrayMax** with Temp	—	✓	✓	1
T/F: comparing two algorithms	✓	—	✓	1
T/F: theoretical analysis	✓	—	✓	1
T/F: the MERGESORT equation	✓	✓	—	2
Hash table with chaining	✓	✓	—	3
Hash table with double hashing	—	✓	—	3
Collision-handling methods and the good hash function	—	—	✓	3
Matrix chain multiplication	✓	✓	—	4+
Longest common subsequence	✓	✓	—	4+
T/F: Huffman, greedy and DP	✓	✓	—	4+
B-tree: build, insert and delete	✓	✓	—	4+
Heaps, BST, trees and graphs	—	—	✓	—

"4+" means a lecture after the third whose slides I have not received. The 442 midterm follows an older, different syllabus.

03 Where are the marks?

The mark distribution of the 451 midterm (out of 25):

Question	Marks	Source
Q1 Matrix chain	4	A lecture after the third.
Q2 Hash table	4	Lecture 3
Q3 Counting operations	2	Lecture 1
Q4 True / False + MCQ	5	One item from Lecture 2, the rest after the third.
Q5 LCS	4	A lecture after the third.
Q6 B-tree	6	A lecture after the third.

Watch out

Lectures 1 to 3 carry about 7 marks out of 25. The rest, about 18 marks, comes from dynamic programming, greedy and B-trees. Do not stop at the first three lectures.

04 Study plan

1 The reliable questions from Lectures 1 to 3

- The **arrayMax** counting table: `7n - 2`, and `7n` with the Temp line.
- Hash tables: chaining and double hashing. Practise with the Practice 2 file.
- The three true/false statements that repeat word for word.

2 The heavy questions

- Matrix chain: learn the cost rule `p·q·r` and fill the table from short chains to long.
- LCS: diagonal + 1 when the letters match, otherwise the larger of top and left.
- B-tree of order 5: split at 5 keys, and borrow or merge on deletion.

They are solved step by step in the 451 midterm solutions file.

3 The last day

Solve the whole 451 midterm against a 90-minute timer, then the mock.